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Incremental Class Detection using Unsupervised Techniques after Outlier Filtering

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Abstract. Accurate classification of leaf diseases in edible plants is essential to maintain healthy agricultural productivity and food security. Recent advances in Convolutional Neural Networks (CNNs) have been proven to be highly effective in classifying leaf diseases in such plants. However, their effectiveness may get challenged while evaluating spurious data in evolving environments. This study introduces an incremental learning approach to classify diseases in *Apple* and *Potato* leaves using CNN for facilitating the preliminary model building stage and then applying clustering techniques to detect outliers. The EfficientNetB0 is utilized here for the initial feature extraction phase, followed by PCA to reduce the number of such features substantially. DBSCAN is used to form representative cluster-centres for each valid leaf-class and isolate outliers within them to improve classifier models. Further, to facilitate detection of outliers during the incremental learning phase, univariate statistical measures of outliers – Z-score and IQR – are employed to determine class-wise feature-wise invalid test-data-points presented to the model. The methodology uses 80% of the labeled dataset for training or pre-learning and the rest for testing or incremental learning. This approach achieved an accuracy of 95.58% for *Apple* and 96.67% for *Potato* leaves, outperforming comparable recent studies.

Keywords: Leaf Disease Classification · EfficientNetB0 · DBSCAN · Incremental Learning · Clustering · Z-Score · IQR · Agricultural Application

1 Introduction

As an agricultural nation, a large segment of the population in India relies heavily on farming as a primary source of income, making it a crucial pillar of the

2 S. Mondal et al.

country's economy. Apple and Potato are staple crops widely consumed across various regions in India. According to the 2021-22 agricultural statistics [1], the combined cultivation area in the whole country for these crops is about 2,202.15 and 308 thousand hectares, producing approximately 53,387.35 and 2,316 thousand tons, respectively. Uttar Pradesh stands out as the top producer of potatoes, trailed by West Bengal and Bihar. In contrast, Himachal Pradesh and Jammu & Kashmir are known for their extensive apple cultivation. Farmers strive to boost crop yields and reduce losses, but unpredictable climate and various plant diseases may pose risks to their profits. Potatoes are very vulnerable to diseases such as 'Early Blight' and 'Late Blight'; these spread quickly and cause premature rotting. In contrast, apples commonly suffer from 'Cedar Rust', 'Black Rot', and 'Scab' diseases, which degrade the quality of the fruit. Traditional disease detection methods are based on farmers' knowledge and experts' inspection, which are time-intensive and costly. Sometimes, experts' timely availability is also an issue in remote agricultural regions. Therefore, developing capable automated systems to detect plant leaf diseases using computer vision techniques can be more effective under such circumstances to mitigate yield losses.

The rest of this paper is organised as follows: Section 1.1 reviews relevant works, Section 1.2 very briefly identifies the gaps present in previous works, Section 1.3 discusses the novelty of the proposed work, Section 2 outlines used methodology, Section 3 analyses experimental outcomes, and Section 4 highlights key findings and future enhancements.

1.1 Related Work

Identifying leaf diseases has become a crucial area of research because of its direct impact on crop productivity, agricultural sustainability, and economic viability. Several techniques for the identification of leaf diseases are documented in the literature [2][3][4], where deep learning methods perform particularly better in classifying image data, as also demonstrated in some of the works reviewed in ensuing paragraphs.

Clustering techniques such as K-Means and Fuzzy C-means (FCM) [5][6][7] are also used as feature extraction tools, while subsequently machine learning algorithms are used for disease classification. Amongst these, in Banana leaf disease detection [5] use of Support Vector Machine (SVM) with shape, color, and texture features achieves an average accuracy of 85%. Similar studies on Grape leaf disease classification [6] using 5,700 preprocessed images utilises grayscale conversion and Laplacian filters with FCM clustering, achieving accuracy close to 95%. K-means clustering is used in [7] as a feature extractor by clustering defective regions in leaf images. Preprocessing through Lab color space and median filtering improves the accuracy further, combined with an SVM classifier.

Deep learning models are mainly used here to address the issues with high precision and scalability. The authors in [8] employ precise and scalable feature extraction and classification schemes to identify various leaf diseases using a multi-layer VGG-19 model. The article [9] illustrates how hybrid unsupervised

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techniques can enhance performance, even in difficult situations, by combining Self-Organized Maps with Deep Boltzmann Machines. The fine-tuned MobileNetV2 [10] achieved an accuracy of 98.27%, whereas the combination of CNN and Random Forest classifier in [11] got 97% accuracy for ‘Mosaic’ disease identification in Apple leaf, showing the benefit of ensemble learning. The authors in [12] used CNN’s extracted features in combination with metaheuristics on the Apple leaf dataset, and their proposed model achieved appreciable accuracy. Transfer learning is applied in [13][14] on pre-trained ResMLP and MobileNet models, respectively. Both fine-tuned models show improved accuracy compared to the base models, and incorporating the attention modules works especially well when labeled data is limited.

1.2 Research Gap Analysis

These studies reflect that CNN models attain impressive accuracy, whereas traditional machine learning models lag a bit behind. However, it may safely be predicted that all models can be further improved by reducing outliers in the data and incorporating incremental learning techniques to enhance performance over time. Outlier detection and handling is a prime factor in this domain, since spurious data (such as an arbitrary leaf image) collected by novice workers may indeed pose serious threat to the classification system otherwise.

1.3 Novelty of the Proposed Work

This study aims to develop an automated system for identifying diseases in Apple and Potato leaves using an incremental learning approach combined with CNN-based feature extraction and unsupervised outlier detection.

The prime contributions of this study are outlined below:

- Usage of DBSCAN for identifying and segregating outliers within each class-labeled data segment after dimensionality reduction using Principal Component Analysis (PCA).
- Calculation of class-wise centroid value to facilitate incremental learning.
- Application of univariate statistical measures of outlier such as Z-Score and Interquartile Range (IQR) on segregated outliers detected by DBSCAN earlier, to supply supportive evidence of outlier range.
- Incorporation of incremental learning strategy during the testing phase with additional benefits of outlier class detection to avoid misclassification of spurious data.

2 Proposed Methodology

The following high level diagram outlines a systematic approach towards classifying leaf diseases by integrating feature extraction through deep learning techniques, followed by outlier detection employing clustering, and lastly incremental learning technique for future improvements of model, as illustrated in Fig. 1.

4 S. Mondal et al.

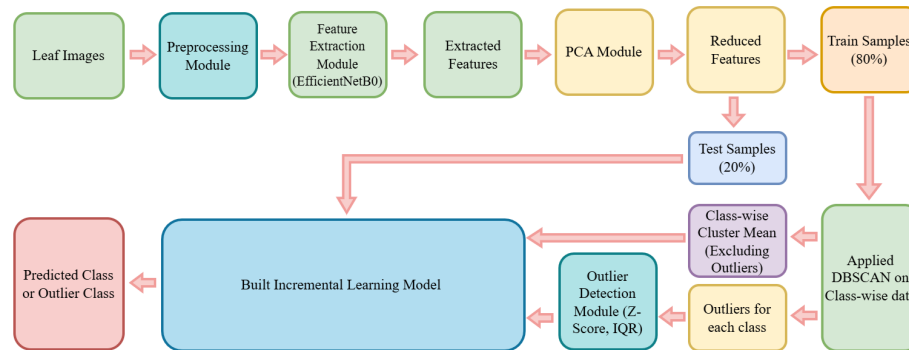


Fig. 1: High Level Diagram of the Proposed Work

2.1 Dataset Description

The used dataset has been collected from a publicly available source [15], which comprises leaf images of 14 different crop species. However, only a few of the crops have sufficient samples for conducting research work. So, only Apple and Potato leaf images have been considered for this research. The Apple dataset includes three disease classes—‘Cedar Rust’, ‘Black Rot’, and ‘Scab’—and a ‘Healthy’ class. Similarly, the Potato dataset consists of two disease classes—‘Late Blight’ and ‘Early Blight’—and a ‘Healthy’ class. A detailed dataset description, including the number of samples per class, is shown in Fig. 2(a).

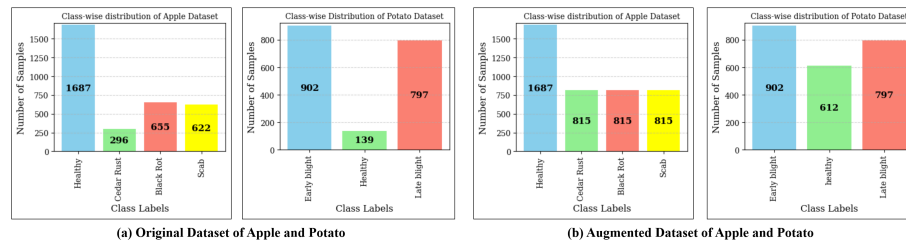


Fig. 2: Dataset Details

2.2 Data Preprocessing and Augmentation

The dataset contains raw images with a resolution ranging from 143×180 to 251×237 with sufficient variation. To standardize the input, all images are resized to 128×128 pixels. As shown in Fig. 2(a), the Apple and Potato datasets have an uneven distribution of samples between categories. Therefore, data augmentation

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techniques from the Keras library [16] are applied during preprocessing to up-sample the under-weighted classes to match the mean sample count in each dataset, limiting synthetic data generation. The details of the augmentation parameters are presented in Table 1 and the class-wise up-sample data is shown in Fig.2(b).

Table 1: Data Augmentation Parameters

Method Name	Default Value	Tuned Value
Rotation Range	0	20
Zoom, Shear Range, Height & Width Shift Range	0.0	0.1
Fill Mode	'constant'	'nearest'
Horizontal Flip	False	True

2.3 Feature Extraction using EfficientNetB0

CNNs are widely recognised as a powerful feature extractor from image data. This study selects EfficientNetB0 as the feature extractor for its superior performance compared to other EfficientNet variants. The main blocks of this model, as described in Fig. 3, consist of Mobile Inverted Bottleneck Convolution (MBConv) and Squeeze-and-Excitation (SE), followed by the Swish activation function. It is designed according to the compound scaling principle (Equations (1), (2), and (3)), which optimally balances depth (d), width (w), and resolution (r) using predefined coefficients to enhance model performance with low computational costs.

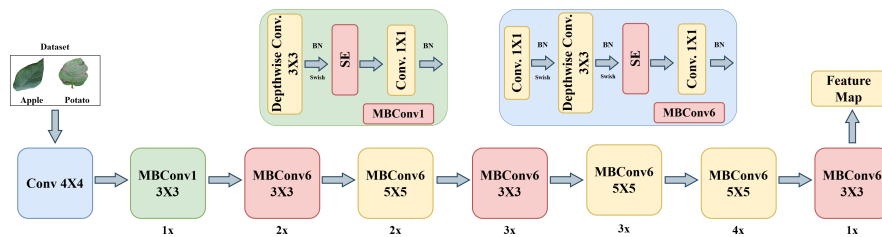


Fig. 3: High Level Diagram of EfficientNetB0

$$d = X^\phi d_0 \quad (1) \quad w = Y^\phi w_0 \quad (2) \quad r = Z^\phi r_0 \quad (3)$$

where X , Y , and Z are coefficients, ϕ is the scaling factor defined by the user, and d_0, w_0, r_0 are the predefined weights of EfficientNetB0.

6 S. Mondal et al.

2.4 Dimensionality Reduction - PCA

The feature set extracted by EfficientNetB0 is represented as $\{f_1, f_2, \dots, f_n\}$. These features exist in a high-dimensional space, and not all of them contribute significantly in the final classification task. The redundant or less informative features increase computational complexity and cause model overfitting. Addressing this problem, **Principal Component Analysis (PCA) is applied to reduce the dimension of the feature space while preserving the most significant information.** The transformed features set $\{F_{\text{PCA}} = f'_1, f'_2, \dots, f'_d\}$ is found by applying the following Equation (4) through PCA, where the original feature size (n) having m samples is denoted as $F(\in \mathbb{R}^{n \times m})$, and $W(\in \mathbb{R}^{d \times n})$ is the projection matrix with d -dimension where $d < n$.

$$F_{\text{PCA}} = W \cdot F \quad (4)$$

2.5 Cluster-Mean Calculation and Outlier Detection

In this study, preliminary outlier detection is performed on training data using the DBSCAN algorithm. DBSCAN utilizes the density distribution technique to identify and separate outliers within a dataset. DBSCAN is an unsupervised clustering algorithm that does not require labeled data. But here, since the dataset used in this study contains labeled instances, DBSCAN is applied separately to each class to compute class-wise cluster centroids $C(\in \{C_1, C_2, \dots, C_N\})$ by the mean of all points after segregating the outliers within each category.

To validate the outliers detected by DBSCAN, statistical methods such as Z-score and IQR are applied. The class-wise Z-score limiting values $Z_{u,c,i}$ and $Z_{l,c,i}$ are set to $+/- \text{Limit}$, where *Limit* is an user-defined constant ranging usually between 2.5 to 3.5 depending on the chosen dataset. Z-scores $Z_{c,i}$ of data points are computed from Equation (5), for the i -th feature of class c , $\mu_{c,i}$ and $\sigma_{c,i}$ being the calculated mean and the standard deviation of that feature $f'_{c,i}$ across m samples of the training dataset.

$$Z_{c,i} = \frac{f'_{c,i} - \mu_{c,i}}{\sigma_{c,i}}, \forall [i \in \{1, \dots, d\}; c \in \{1, \dots, k\}] \quad (5)$$

Similarly, the IQR method identifies class-wise outliers by calculating the range between the first quartile ($Q1_{c,i}$) and the third quartile ($Q3_{c,i}$), Equation (6) represents the class-wise lower range $IQR_{l,c,i}$ and Equation (7) upper range $IQR_{u,c,i}$; values beyond this range are considered potential outliers.

$$IQR_{l,c,i} = Q1_{c,i} - 1.5 \times Z \quad (6) \quad IQR_{u,c,i} = Q3_{c,i} + 1.5 \times Z \quad (7) \quad Z = (Q3_{c,i} - Q1_{c,i}) \quad (8)$$

The outlier detection threshold for each class is then determined using Equations (5), (6), (7), and (8) as outlined in Algorithm 1. The inputs in Algorithm 1 include the set of identified outliers by DBSCAN as $O = \{O_1, O_2, \dots, O_n\}$, along with the reduced feature set F_{PCA} and an user-defined choice of P value for further reduction in feature set count. This approach ensures that only the

Title Suppressed Due to Excessive Length 7

crucial features will participate in finding outliers at the time of testing, thus improving the model's Time-complexity.

Algorithm 1 Class-wise Outlier Detection

Input: O , F_{PCA} and P .

Output: Selected features F_{final} for outlier detection.

- 1: Initialize $M[d] \leftarrow N[d] \leftarrow 0$
 - 2: **for** each outlier $O_k \in O$ and $f'_i \in F_{PCA}$ **do**
 - 3: Add 1 to M_i if value(f'_i) lies outside range $Z_{u,c,i}$ and $Z_{l,c,i}$ else Add 0 to M_i .
 - 4: Add 1 to N_i if value(f'_i) lies outside range $IQR_{u,c,i}$ and $IQR_{l,c,i}$ else Add 0 to N_i .
 - 5: **end for**
 - 6: Sort $\downarrow$ on $M[d]$ and $N[d]$ values, maintaining proper mapping onto F_{PCA} .
 - 7: Select top P features from *both sets*.
 - 8: Compute the intersection between M and N : $F_{final} \leftarrow M \cap N$
-

2.6 Incremental Learning-based Classification

The proposed incremental learning technique adapts itself to new data. Initially, the model is built on class-wise samples from 80% of the F_{PCA} (D_{train}). The remaining 20% of data is kept for testing (D_{test}). For each sample T_i from D_{test} , first apply the proposed class-wise outlier detection module (Section 2.5). If T_i is identified as an outlier in all classes, it is declared as an outlier class. However, if T_i is an outlier in the $N - 1$ clusters having centroids $C (\in \{C_1, C_2, \dots, C_N\})$, then it's predicted label gets updated as the label of the remaining cluster. Otherwise, the predicted class label of T_i is determined through the label of the particular cluster-mean among all non-outlier cluster centroids $C' \subseteq C$ that has the minimum Euclidean distance D to T_i . Therefore, the cluster centroid is updated by averaging all points in the cluster, including the new sample. Thus, the model will dynamically calibrate itself with the new data. This process repeats until all test samples have been considered, as described in Algorithm 2.

3 Results and Discussion

The experimental setup details and results are presented in the following subsections.

3.1 Experimental setup

All experiments in this study utilised an 8 GB variant of a Windows 11 PC with an 11th-gen Core i3 CPU and 256 GB SSD. CNN models have been trained with L4/T4 GPUs in the Google Colaboratory with 32 GB RAM. Unsupervised machine learning and statistical techniques of outlier detection such as DBSCAN, Z-score and IQR have been adapted to incorporate Incremental Learning Strategies.

8 S. Mondal et al.

Algorithm 2 Incremental Learning-Based Classification**Input:** D_{test} , Cluster centroids $C \in \{C_1, C_2, \dots, C_N\}$, and Selected features F_{final} **Output:** Predicted class as either Outlier or $C_{\text{Predicted}} \forall D_{\text{test}}$

```

1: for each sample  $T_i \in D_{\text{test}}$  do
2:   Initialize outlier flag array:  $A \leftarrow [0, 0, \dots, 0] \in \mathbb{R}^N$ 
3:   for each cluster  $x_i$  having centroids  $\in C$  do
4:     if  $T_i$  is an outlier for cluster  $x_i$  using  $F_{\text{final}}$  then Set  $A[x_i] \leftarrow 1$ 
5:     end if
6:   end for
7:   if  $\forall j, A[j]$  is 1 then
8:     Declare  $T_i$  as an outlier
9:     break
10:  else if  $\exists! j : A[j]$  is 0 then
11:     $C_{\text{Predicted}} \leftarrow \text{Label of } C[j]$ 
12:  else
13:    Identify all non-outlier cluster means as  $C'$  ( $C' \subseteq C$ ) where  $\forall j, A[j]$  is 0
14:    Compute  $C^* \leftarrow \arg \min_{C'} \text{distance}(T_i, C') : C_{\text{Predicted}} \leftarrow \text{Label of } C^*$ 
15:  end if
16:  Update  $C[C_{\text{Predicted}}]$  by recalculating the mean including  $T_i$ 
17: end for

```

3.2 Results - Features Reduction and Selection

Principal Component Analysis (PCA) reduces the dimensionality of features extracted using EfficientNetB0 for both datasets. The method retains critical attributes by setting the variance level at 95%, significantly reducing the feature space. The detailed feature reductions for each dataset are outlined in Table 2.

Table 2: PCA-Based Feature Reduction

Dataset	Total No. of Samples	No. of Features	Reduced Features by PCA
Apple	4132	1280	279
Potato	2311	1280	239

3.3 Results - Cluster-Mean and Outlier Detection

The experimental results of detecting outliers from D_{train} using DBSCAN are visualized in Fig. 4 for the Apple leaf disease classes, where separate colors (*Blue*, *Red*, *Cyan* and *Green*) represent distinct classes, and outliers are highlighted in *Yellow*. The Apple dataset contains 279 features; for visualization, two representative features were selected. For each class, the cluster mean is calculated twice: once including all data points (Fig. 4(a,b,c,d)), and again after removing the identified outliers (Fig. 4(e,f,g,h)). A similar experiment was

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conducted on the Potato leaf disease dataset, with the results retained for subsequent analysis. The optimal hyperparameters—EPS (neighborhood radius) and MinPts (minimum neighbors)—were determined using the elbow method to balance cluster coherence and outlier detection. Table 3 summarizes the derived hyper-parameters, including class-wise outliers and detection time.

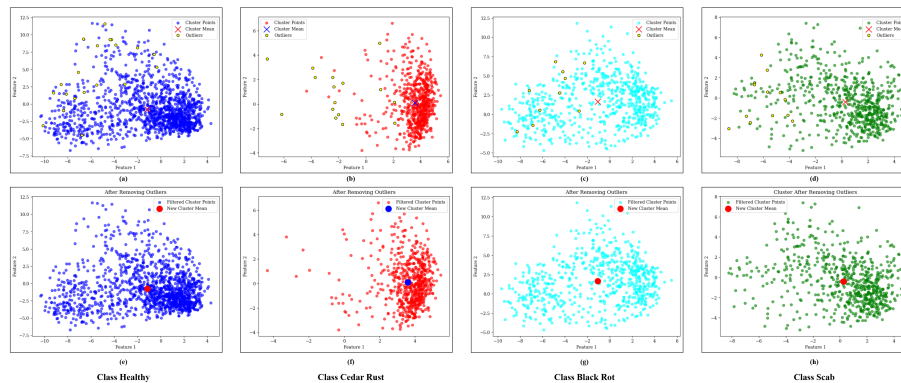


Fig. 4: Class-wise Outliers of Apple Classes on Reduced Features

Table 3: Hyper-parameters of DBSCAN including Outlier Detection Summary

Dataset	Class	EPS	MinPts	No. of Outliers	Time Taken (sec.)
Apple	Healthy	10.25	5	20	0.032
	Cedar Rust	8.6	5	19	0.012
	Black Rot	11.82	5	13	0.013
	Scab	11.07	5	11	0.013
Potato	Healthy	11.51	5	27	0.009
	Early Blight	9.57	5	20	0.011
	Late Blight	10.39	5	11	0.014

Additional statistical outlier detection techniques, such as the Z-score and Interquartile Range (IQR), were used to further validate the outliers detected by DBSCAN. The results confirm that the outliers identified by DBSCAN aligned with those detected using the Z-score and IQR. However, these statistical methods perform univariate analysis, so they identified additional outliers that DBSCAN did not detect. Given that the dataset is derived from image features, consisting of 279 and 239 high-dimensional features of Apple and Potato, respectively, variations in the detection results were expected.

Furthermore, a threshold range for outlier elimination was calculated for each class (Algorithm 1) based on the combined findings of DBSCAN, Z-Score, and IQR. This threshold is a reference during the testing phase, ensuring that outliers are separated accordingly. Since outlier thresholds are in P -dimension and also high-dimension in nature, only the ranges of the top 5 features are

10 S. Mondal et al.

reported for each class in Table 4. The calculated threshold filters out noisy data and promotes consistent classification performance by effectively identifying the outlier class.

Table 4: Top 5 Feature-wise Outlier Boundaries for Different Classes

Dataset	Class Name	Bound	F1	F2	F3	F4	F5
Apple	Healthy	Lower	-9.93	-0.69	-0.43	-0.43	-0.41
		Upper	8.79	0.7	0.43	0.42	0.4
	Cedar Rust	Lower	-1.19	-0.38	-0.38	-0.63	-0.56
		Upper	8.14	0.38	0.37	0.63	0.61
	Black Rot	Lower	-0.85	-7.13	-0.52	-0.52	-0.86
		Upper	0.85	10.3	0.51	0.54	0.83
	Scab	Lower	-4.05	-0.74	-0.75	-3.32	-0.45
		Upper	3.37	0.76	0.7	2.93	0.47
Potato	Early Blight	Lower	-2.93	-2.02	-1.68	-1.58	-1.43
		Upper	2.95	1.99	1.72	1.61	1.39
	Healthy	Lower	-6.08	-0.66	-0.66	-0.61	-0.65
		Upper	7.8	0.66	0.67	0.64	0.65
	Late Blight	Lower	-6.52	-0.71	-0.69	-0.7	-0.68
		Upper	8.41	0.69	0.72	0.7	0.66

3.4 Results - Incremental Learning-based Classification

Incremental learning was evaluated on the Apple and Potato datasets, which consist of 4,132 images across 4 classes and 1,848 images across 3 classes, respectively. For performance assessment, 827 samples from the Apple dataset and 463 from the Potato dataset were designated as the test set (D_{test}).

During the incremental learning phase, the model used reduced features (F_{PCA}) of dimension d by PCA, for predicting the class value. Whereas, outlier detection was subsequently applied to the reduced features (F_{final} in Algorithm 1) having dimension P , where $P < d$, as detailed in Section 3.3. In this study, the P -value of 66 for Apple and 68 for Potato was used, resulting in the identification of 16 and 27 outliers, respectively. After removing the detected outliers, the model achieved classification accuracies of 95.58% for Apple and 96.67% for Potato, as summarized in Table 5. Moreover, the dimensionality reduction used in the outlier detection phase contributed to a reduction in computational complexity for both datasets. Importantly, the model demonstrated consistent performance across both datasets, despite differences in data size and class distribution. This indicates the adaptability and generalizability of the proposed incremental learning model in diverse scenarios.

3.5 Comparison with Related Works

The accuracy obtained through incremental learning exceeds that of the latest state-of-the-art models. A comparative analysis was performed against the most recent relevant studies published between 2024 and 2025, with the results summarised in Table 6.

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Table 5: Comparison of Metrics in Incremental Learning

Category	Apple	Potato
Total Samples	4132	2311
Test Samples (D_{test})	827	463
Detected Outliers on D_{test} using F_{final}	16	27
Accuracy % on D_{test} after Outlier Removal	95.58	96.67

Table 6: Comparison of Proposed Approach with Existing Works

Ref	Year	Category	Used Method	Outlier Detection	Acc (%)
[4]	2024	Apple	VGG19	×	94.7
[2]	2024		DenseNet201	×	95.32
[17]	2024		BLSENet	×	93.58
[20]	2023	Potato	VGG16	×	95.36
[18]	2024		Non-Linear SVM	×	95.6
[19]	2024		CNN	×	92
			<i>Proposed Work</i>		
		Apple	Incremental Learning	✓	95.58
		Potato	Incremental Learning	✓	96.67

4 Conclusions and Future Work

This study integrates an incremental learning model with outlier detection techniques to eliminate noise from the dataset, ensuring better segregation of class data progressively over time. Since this learning strategy involves continuous adaptation, removal of faulty test samples is essential; else, the model would learn from errors and degrade in performance. The excessive time-complexity involved in determining class values is somewhat mitigated in this approach by comparing valid test instances with only the class-wise centroid instead of all individual data points within the class. Dimensionality issues during such comparisons are further reduced by applying PCA at the outset. **The proposed method thus achieves an accuracy of 95.58% on the Apple dataset and 96.67% on the Potato dataset, demonstrating its effectiveness over earlier related works employing deep learning and machine learning techniques. Further time-complexity reduction is also achieved here by enhancing the outlier detection process using some time-tested statistical measures such as Z-score and IQR strategies. These serve dual purpose: first, that of setting up a normal value range for each PCA feature; second, finding the most influential feature subset which contributes towards the class-wise outlier determination during the training phase using DBSCAN.**

For future research, the framework can be extended to diverse agricultural datasets with larger sample sizes, allowing for more refined and feature-specific clustering. Additionally, the current study utilises CNN-based feature extraction, which can be substituted with color-based feature extraction techniques to capture essential attributes and reduce feature dimensionality. Enhancing the

12 S. Mondal et al.

model by combining varied datasets and different feature extraction techniques can optimize scalability and adaptability in diverse agricultural applications.

References

1. Department of Agriculture & Farmers Welfare: *Agricultural Statistics at a Glance 2022*. Government of India (2022). Available at: <https://desagri.gov.in/wp-content/uploads/2023/05/Agricultural-Statistics-at-a-Glance-2022.pdf> (Accessed: 2 April 2025)
2. Parihar, A., Batra, S., Mahajan, M.: *Deep Learning-based Models for Apple Leaf Disease Diagnosis: Implementation and Comparative Performance Assessment*. In: 2024 8th International Conference on Electronics, Communication and Aerospace Technology (ICECA), pp. 1098–1104, IEEE (2024).
3. Kumar, S., Kumar, R., Gupta, M., Obaid, A.J.: *Deep Learning-Based Web Application for Real-Time Apple Leaf Disease Detection and Classification*. In: 2024 International Conference on Advancements in Smart, Secure and Intelligent Computing (ASSIC), pp. 1–6, IEEE (2024).
4. Rajeshwar, M., Das, B., Sowjanya, K.: *A CNN Model For Plant Leaf Disease Detection*. In: 2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT), pp. 1–6, IEEE (2024).
5. Chaudhari, V., Patil, M.: *Banana leaf disease detection using K-means clustering and Feature extraction techniques*. In: 2020 International Conference on Advances in Computing, Communication & Materials (ICACCM), pp. 126–130, IEEE (2020).
6. Chaimanee, A., Fu, X., Lyu, G., Angkawisittpan, N., Chaiduangri, N., Sultornsanee, S.: *Improved Fuzzy C-means Clustering for Grape Leaf Disease Detection using Mathematical Morphology*. In: 2023 International Electrical Engineering Congress (IEEECON), pp. 391–394, IEEE (2023).
7. Rani, F.A.P., Kumar, S.N., Fred, A.L., Dyson, C., Suresh, V., Jeba, P.S.: *K-means Clustering and SVM for Plant Leaf Disease Detection and Classification*. In: 2019 International Conference on Recent Advances in Energy-efficient Computing and Communication (ICRAECC), pp. 1–4, IEEE (2019).
8. Rajeshwar, M., Das, B., Sowjanya, K.: *A CNN Model For Plant Leaf Disease Detection*. In: 2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT), pp. 1–6, IEEE (2024). <https://doi.org/10.1109/ICCCNT61001.2024.10724786>
9. Janani, V., et al.: *A Novel Approach for Detecting Apple and Maize Plant Diseases Using Self Organized Maps and Deep Boltzmann Machine*. In: 2024 Second International Conference on Advances in Information Technology (ICAIT), pp. 1–6, IEEE (2024)
10. Belmir, M., Difallah, W., Ghazli, A.: *A Reliable Apple Leaf Disease Identification Using a Deep Learning-Based MobileNetV2 to Safeguard Apple Fruit Safety*. In: 2024 4th International Conference on Embedded & Distributed Systems (EDIS), pp. 279–284, IEEE (2024)
11. Kaur, A., Kukreja, V., Aggarwal, P., Thapliyal, S., Sharma, R.: *Amplifying Apple Mosaic Illness Detection: Combining CNN and Random Forest Models*. In: 2024 IEEE International Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI), pp. 1–5, IEEE (2024)
12. Tajji, K., Sohail, A., Shahzad, T., Shoaib Khan, B., Adnan Khan, M., Ouahada, K.: *An Ensemble Hybrid Framework: A Comparative Analysis of Metaheuristic Algorithms for Ensemble Hybrid CNN Features for Plants Disease Classification*. IEEE Access, 12, 61886–61906 (2024).
13. Srivastav, S., Guleria, K., Sharma, S., Singh, G.: *Apple Leaf Disease Detection using Deep Learning Based Pre-Trained Models*. In: 2024 4th International Conference on Artificial Intelligence and Signal Processing (AISP), pp. 1–5, IEEE (2024).
14. Srivastav, S., Guleria, K., Sharma, S.: *Apple Leaf Disease Detection using Deep Learning-based Convolutional Neural Network*. In: 2024 4th Asian Conference on Innovation in Technology (ASIANCON), pp. 1–5, IEEE (2024).
15. Moupojou, E.: *PlantVillage without background [Data set]*. Zenodo (2023). <https://doi.org/10.5281/zenodo.3163800>. Accessed on January 9, 2025.
16. TensorFlow, *ImageDataGenerator*. TensorFlow API Documentation. Available: https://www.tensorflow.org/api_docs/python/tf/keras/preprocessing/image/ImageDataGenerator.
17. T. Fang, J. Zhang, D. Qi and M. Gao, BLSENet: A novel lightweight bilinear convolutional neural network based on attention mechanism and feature fusion strategy for apple leaf disease classification, *Journal of Food Quality*, 2024 (2024) 5561625.
18. Srivastava, A., Singh Rawat, B., Sawan, V., Vishal, and Dhondiyal, S.A.: *Potato Leaf Disease Detection Method Based on Support Vector Machines*. In: Proceedings of the 2024 Second International Conference on Advanced Computing & Communication Technologies (ICACCTech), pp. 711–715. IEEE (2024).
19. Rozaqi, A.J., and Sunyoto, A.: *Identification of Disease in Potato Leaves Using Convolutional Neural Network (CNN) Algorithm*. In: Proceedings of the 2020 3rd International Conference on Information and Communications Technology (ICOIACIT), pp. 72–76. IEEE (2020).
20. Shiffa, T.S., Suchithra, M.S., and Vijayakumar, A.: *Potato Leaf Diseases Detection Using Machine Learning and Deep Learning*. In: Proceedings of the 2023 Intelligent Computing and Control for Engineering and Business Systems (ICCEBS), pp. 1–6. IEEE (2023).